



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This project focuses on the design and implementation of an autopilot system for the longitudinal dynamics of an aircraft, with the goal of achieving control and stability over vertical speed and pitch rate. The primary objectives include developing a vertical speed guidance loop, integrated with an inner pitch rate control loop, and an auto-throttle system to regulate airspeed. Through theoretical analysis and simulation, the system's performance was evaluated based on key criteria, including response time, stability, and robustness to external disturbances. A 6DOF nonlinear simulation was used to validate the control laws, demonstrating the autopilot's ability to track vertical speed and airspeed commands with minimal overshoot and fast settling times. The system's response to gust inputs was also assessed to evaluate its disturbance rejection capabilities. The results confirm that the proposed longitudinal autopilot design meets the specified performance requirements, providing a reliable solution for maintaining desired flight paths and stability in the longitudinal plane.

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1. Introduction

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2. Q1 – Vertical Speed Transfer Function

To assess the feasibility of using the aircraft elevator (δ_e) to directly control the vertical speed (v_s), the transfer function $G_{\delta_e}^{v_s}$ was determined. Where $C = [0 \quad -1 \quad 0 \quad V_{trim}]$, the Transfer Function for v_s was found to be (2.1).

$$G_{\delta_e}^{v_s}(s) = \frac{8.57(s - 16.89)(+12.79)(s + 0.04162)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (2.1)$$

The stability of aircraft's control system can be determined by the location of its poles, shown in Figure 1. Poles can be seen on the right half of the s-plane, demonstrating instability.

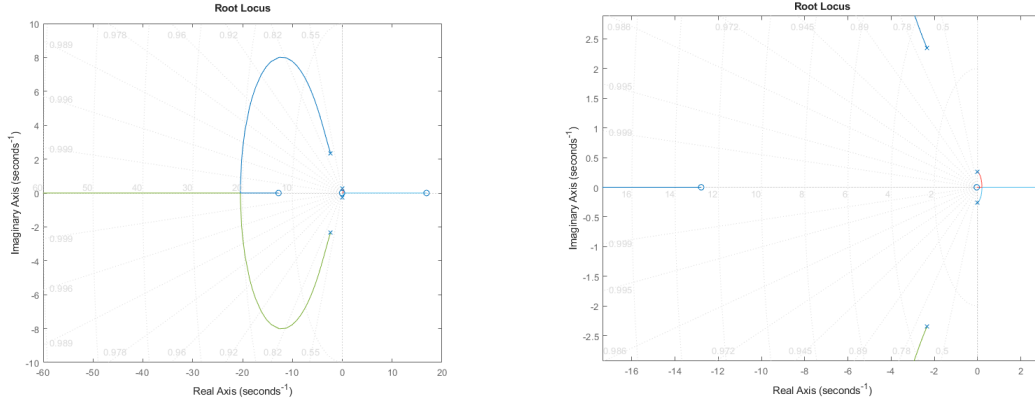


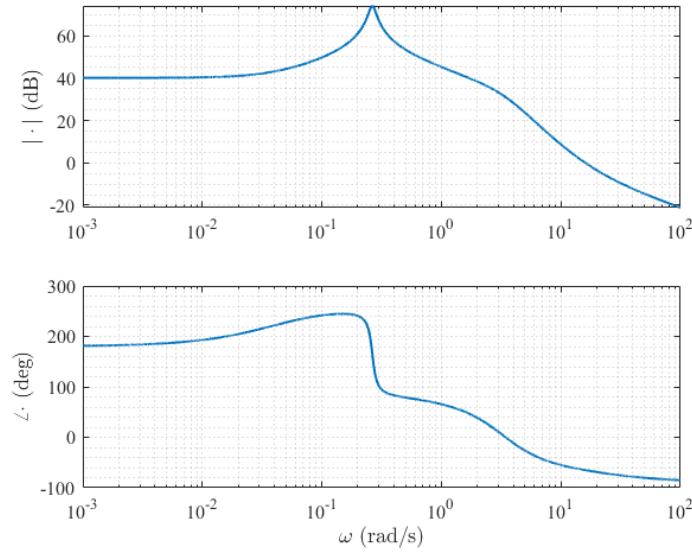
Figure 1: Root Locus of $G_{\delta_e}^{v_s}(s)$

Further, when analysing the Bode Plot of $G_{\delta_e}^{v_s}(s)$ in Figure 2, a resonant peak at a low frequency can be seen, indicating high sensitivity and possible instability. The phase response can be lagging at low frequencies. Thus, designing an autopilot using elevator to directly control vertical speed would be challenging due to the indirect relationship between elevator input and vertical speed output.

3. Q2 – Pitch Rate Autopilot

the following is for my understanding -

The primary function of the elevator is to control the pitch, which influences the angle of attack, which then in turn affects the lift produced by the wings. When the pitch angle is altered, there is a delayed response as the aircraft's trajectory adjusts to the new wing orientation. This is because the resultant lift force must change sufficiently to alter to the vertical component of the aircraft's velocity.

Figure 2: $G_{\delta_e}^{v_s}(s)$ Bode Plot

4. Q3 – Vertical Speed Guidance Loop

The vertical speed guidance loop is designed to control the aircraft's ascent and descent rates by incorporating a pitch rate controller as an inner loop. This loop includes a compensator in the forward leg, which processes the vertical speed error to generate a pitch rate command. The pitch rate command then feeds into the inner loop to modulate the elevator input, allowing the system to adjust vertical speed in response to deviations from the desired reference.

4.1. Loop Topology

The complete loop topology of the vertical speed guidance loop is shown in Figure 3. The compensator transforms the vertical speed error ($vs_{ref} - vs$) into a pitch rate command. This command is then processed within the inner loop section ($de2q$), where the transfer functions G_1 and G_2 are used in the pitch rate controller to compute the required elevator deflection (δ_e) to achieve the target pitch rate.

4.2. Aircraft Response to Vertical Speed Command

The vertical speed response, in Figure 4, demonstrates the ability for the system's response to a 500 ft/min (2.54 m/s) vertical speed step input command.

The vertical speed response, shown in Figure 2, demonstrates the system's performance in tracking a 500 ft/min (approximately 2.54 m/s) vertical speed command step input. The response shows a rapid increase in vertical speed, reaching close to the desired 2.5 m/s target within the first 2 seconds. There is a slight overshoot beyond the target speed, followed by a gradual settling towards the reference value, achieving stability within approximately 10 seconds. This response indicates that the system meets the rise time and settling time requirements, effectively balancing speed of response

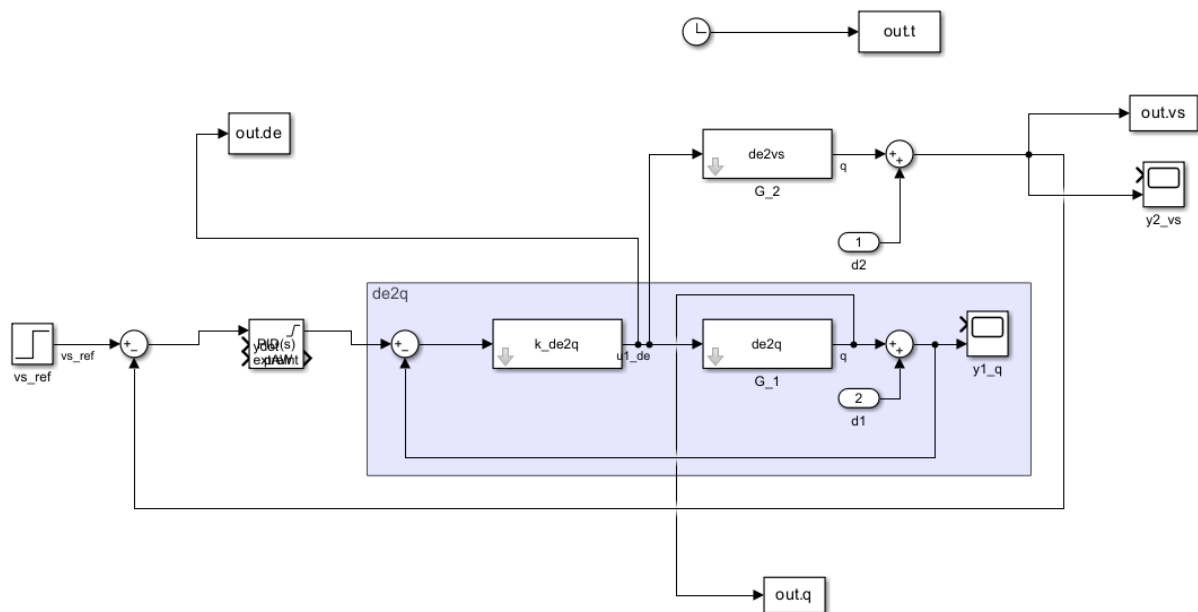


Figure 3: Vertical Speed Guidance Loop Topology

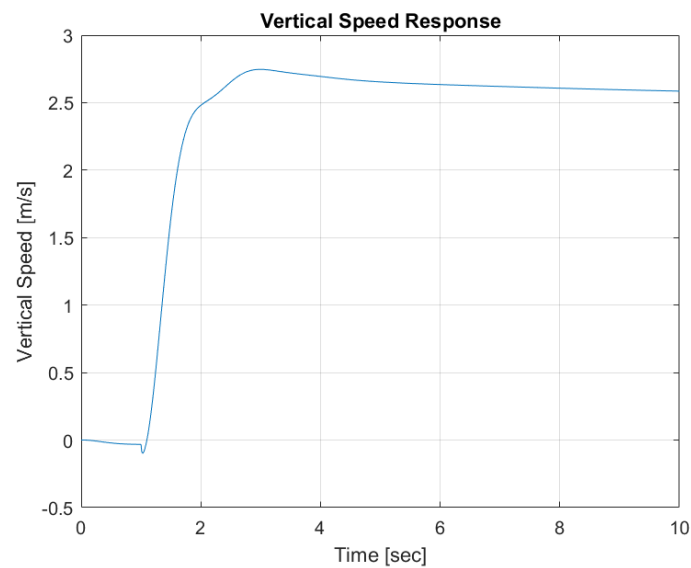


Figure 4: Vertical Speed Response

with controlled damping. The smooth convergence to the command confirms the system's capability to achieve accurate vertical speed tracking.

4.3. Pitch Rate Response

The pitch rate response, shown in Figure 5, reflects the performance of the inner pitch rate controller.

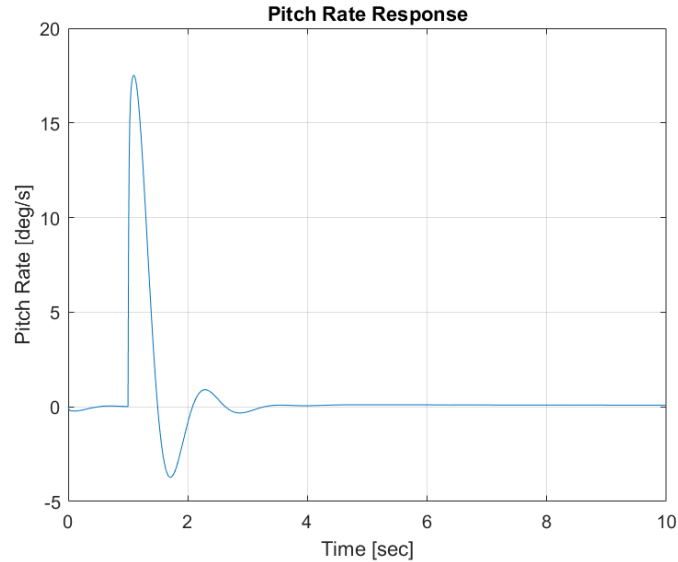


Figure 5: Pitch Rate Response

The response shows a peak reaching 16deg/s , followed by a quick correction with minor oscillations before settling to zero. This peak and correction indicate the controller's rapid adjustment to the pitch rate required to meet the vertical speed command of 500ft/min (2.54m/s), as outlined in Section 4.2. These damped oscillations demonstrate the pitch rate controller's stability and responsiveness.

4.4. Elevator Activity

Figure 6 displays the control activity of the elevator in response to the inputted vertical speed command.

The elevator initially deflects sharply to approximately -2.3degrees as the controller adjusts to achieve vertical speed command, followed by the elevator deflection stabilizing back to zero, demonstrating its minimal effort required once the target vertical speed is reached.

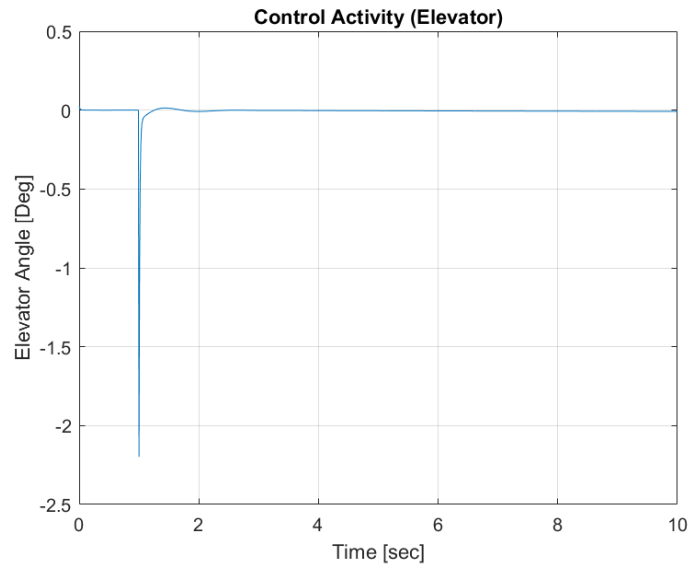


Figure 6: Elevator Control Activity

5. Q4 – Design Auto-Throttle Control System

In this section an auto-throttle design was attempted, wherein errors in airspeed would determine throttle command.

5.1. MIMO Loop Topology

The auto-throttle system was to be achieved with the MIMO loop topology shown in Figure 7.

5.2. Guidance Loop Topology Specifications

This loop was designed to the same specifications as the vertical speed guidance loop, with a vertical speed step input of 500 ft/min. The other constraints could not be implemented however due to difficulties in controlling the system. In addition to the vertical speed input there was also an input for u (airspeed).

5.3. Primary Control Effects

The primary control effects, namely the transfer function of throttle to airspeed ($(dt2u)$), and elevator to vertical speed ($de2vs$), are shown in Figures 8 and 9.

These effects tend to follow a more sinusoidal pattern, reaching similar magnitudes over the simulation period. This is quite different from the responses found from solely the vertical speed control loop (in 4.1). The throttle is more effective in making airspeed changes because of its direct affect on this output, coupled with the effects from the elevator control, although because of the increased complexity of the system it is more difficult to tune, and does not reach a stable point (like the outputs of the vertical speed control loop).

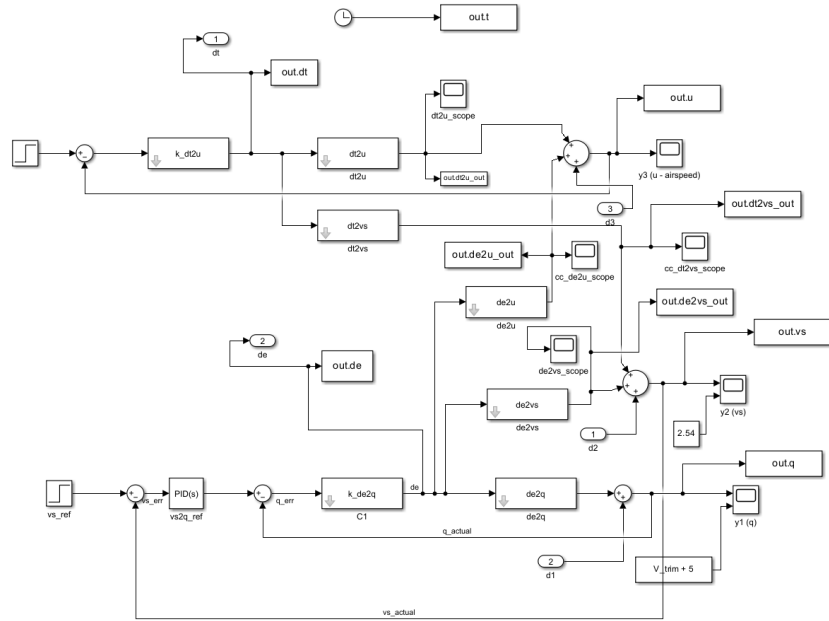


Figure 7: MIMO Simulink Loop for Auto-Throttle Design

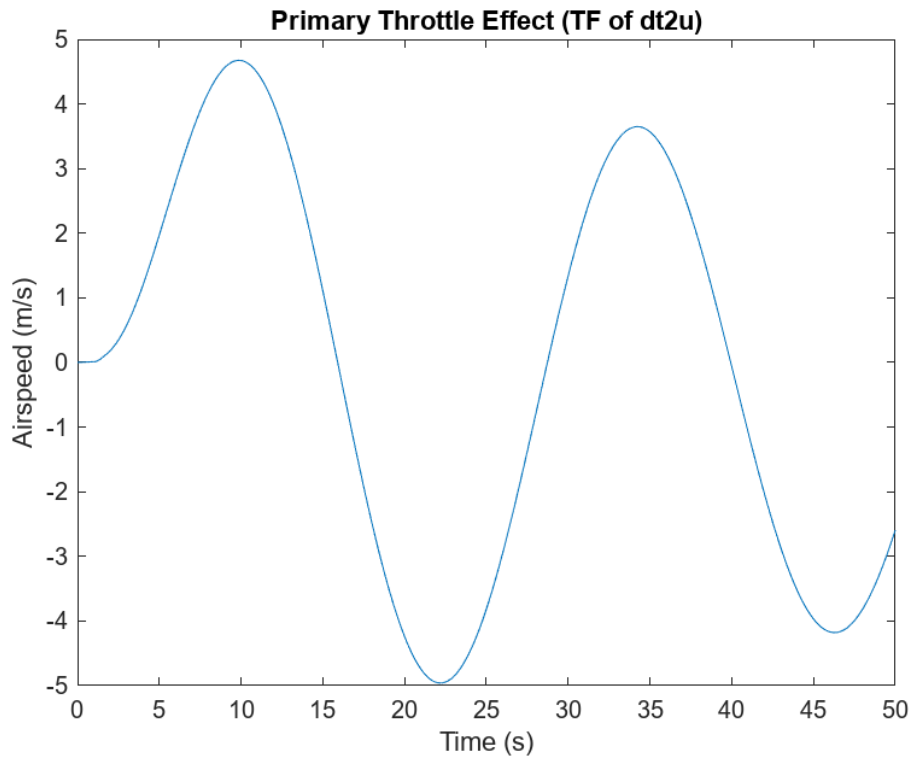


Figure 8: Primary Control Effect: Throttle to Airspeed

The overall output of vertical speed and pitch rate (from both elevator and throttle control) however show a more similar trend to the vertical speed control loop, shown in Figures 10 and 11, albeit with a

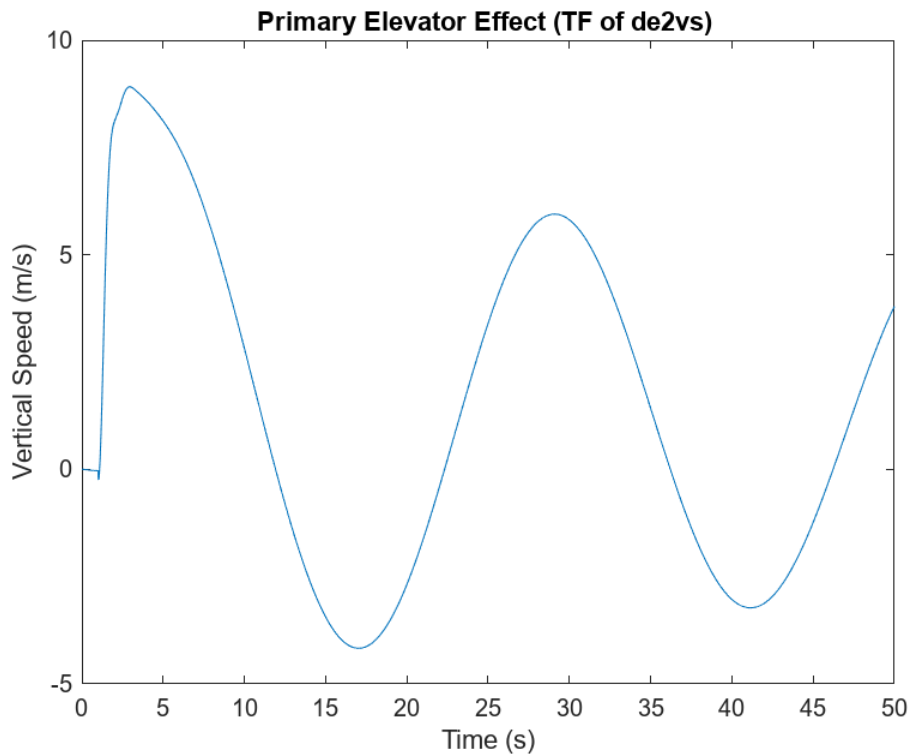


Figure 9: Primary Control Effect: Elevator to Vertical Speed

longer settling time (i.e., not being controlled in the simulation time). Airspeed will not be compared as it is not explored in the vertical speed control loop. However it is included for completeness (COME BACK TO THIS).

5.4. Cross-coupling Control Effects

The cross-coupling control effects, namely the transfer function of throttle to vertical speed ($(dt2vs)$), and elevator to airspeed ($de2u$), are shown in 12 and 13.

Again, compared to only the vertical speed control loop, these responses are sinusoidal, and do not stabilise in the timeframe - showing the complexities of cross looping.

5.5. Elevator and Throttle Response

The control activities of the elevator and throttle in response to the inputted vertical speed and airspeed commands are shown in Figures 14 and 15. The throttle follows a reasonable trend, appearing to begin stabilising at 50 seconds in response to the airspeed step input (although not having a stabilising effect on airspeed as seen in Figure 8). The elevator seems to be more erratic, with a large initial deflection, and then no change for the simulation period. This may again be due to the cross coupling effects, and the increased complexity of the system.

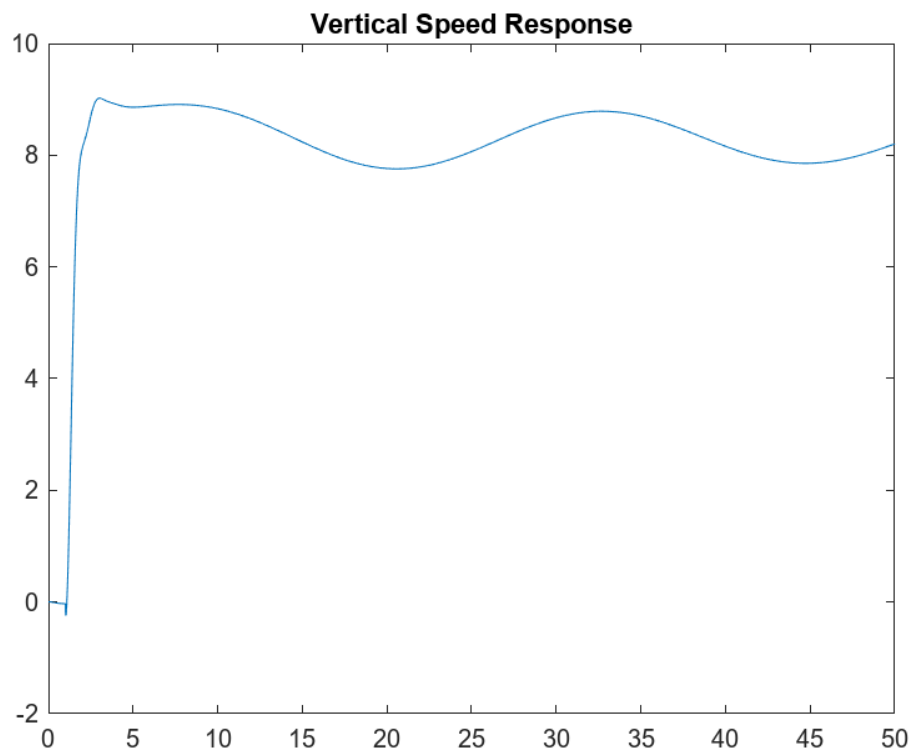


Figure 10: Vertical Speed Response (Auto Throttle System)

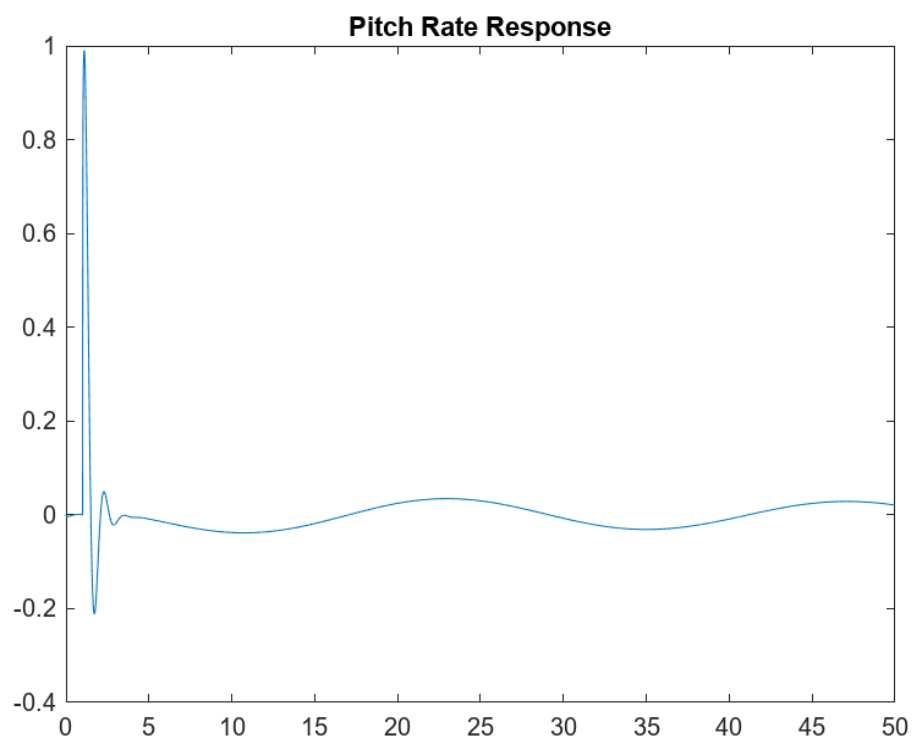


Figure 11: Pitch Rate Response (Auto Throttle System)

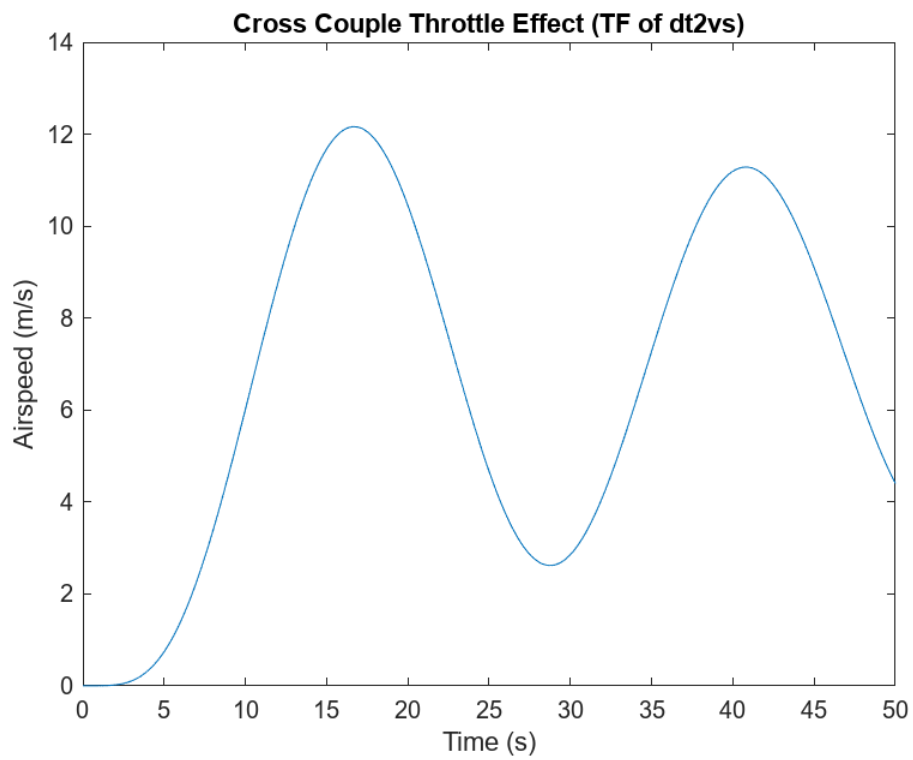


Figure 12: Cross Couple Response (Throttle to Vertical Speed)

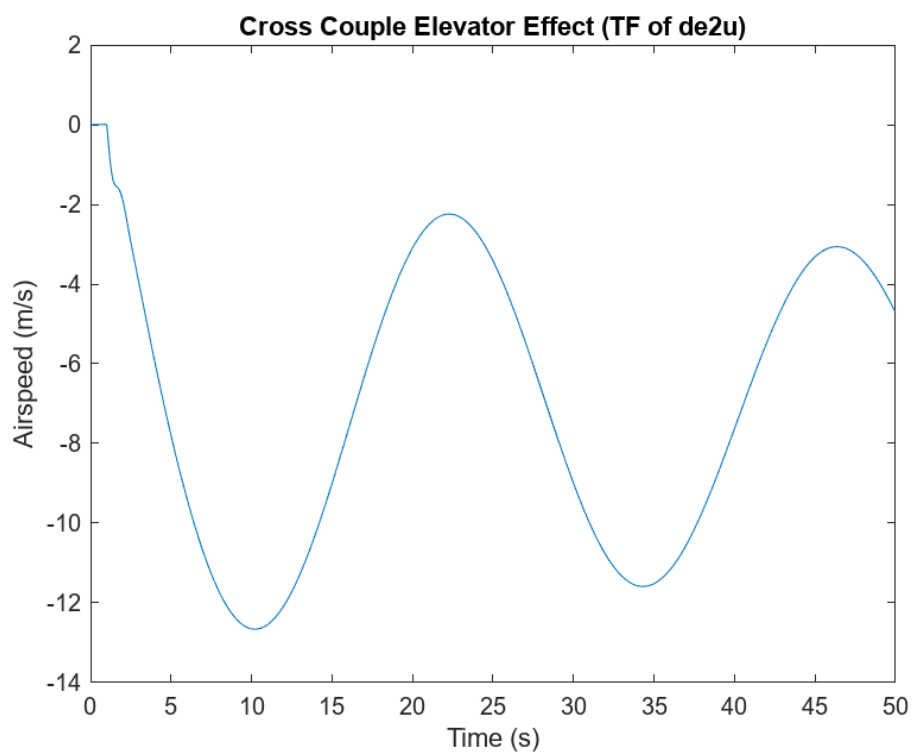


Figure 13: Cross Couple Response (Elevator to Airspeed)

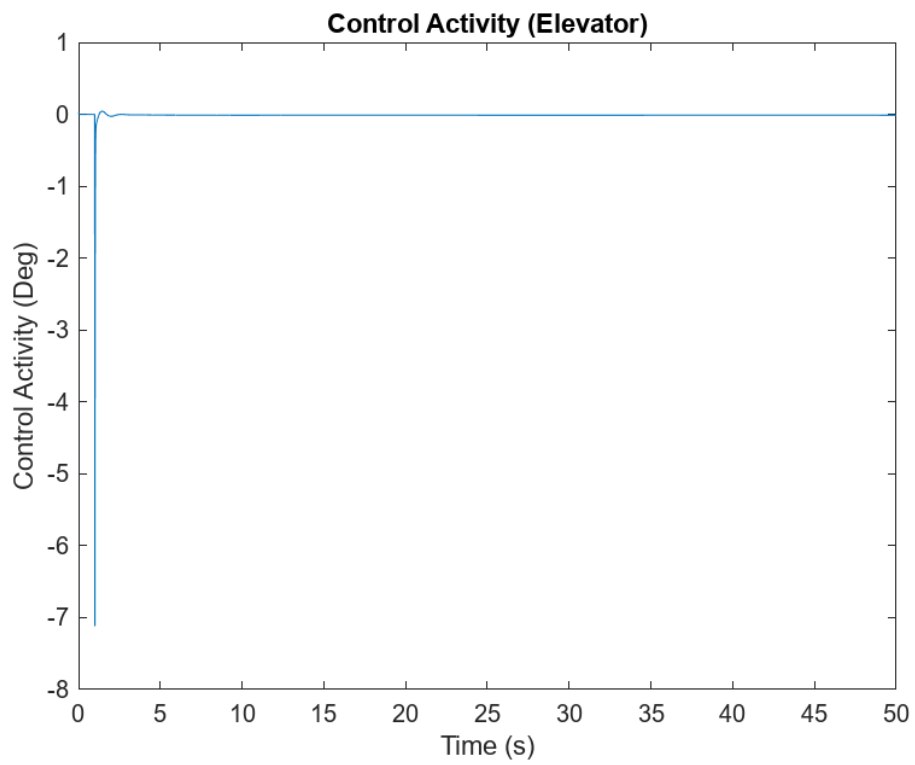


Figure 14: Elevator Control Activity (Auto Throttle System)

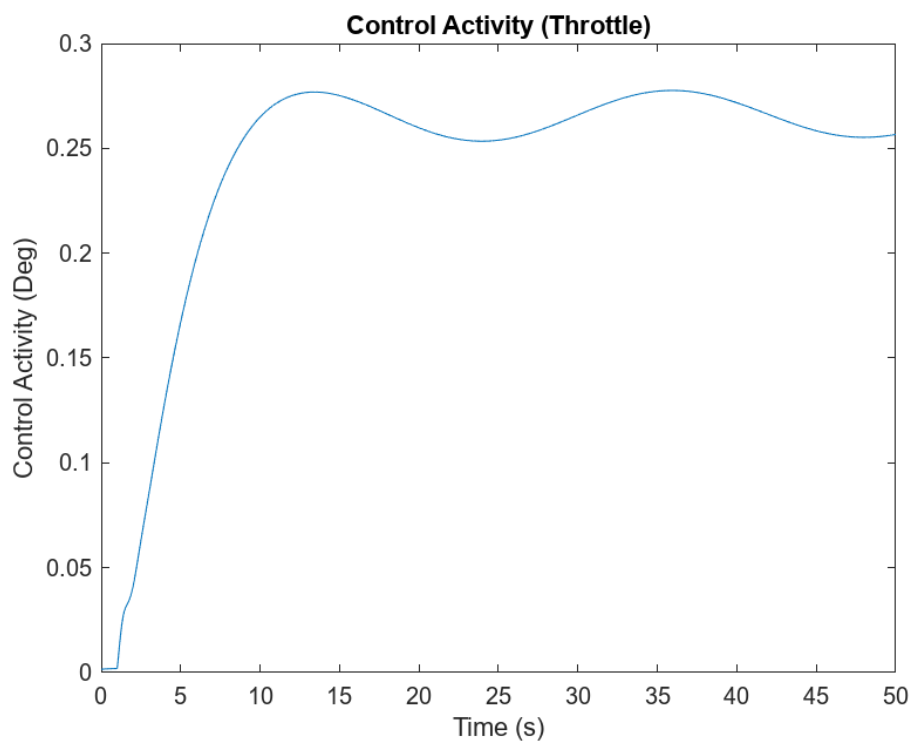


Figure 15: Throttle Control Activity (Auto Throttle System)

5.6. Auto-Throttle Effectiveness

The effectiveness of the auto throttle system without the vertical speed controller was explored, both with and without the pitch rate controller. It was discovered that the vertical speed controller is necessary for non-erratic behaviour of airspeed, as seen in Figure 16.

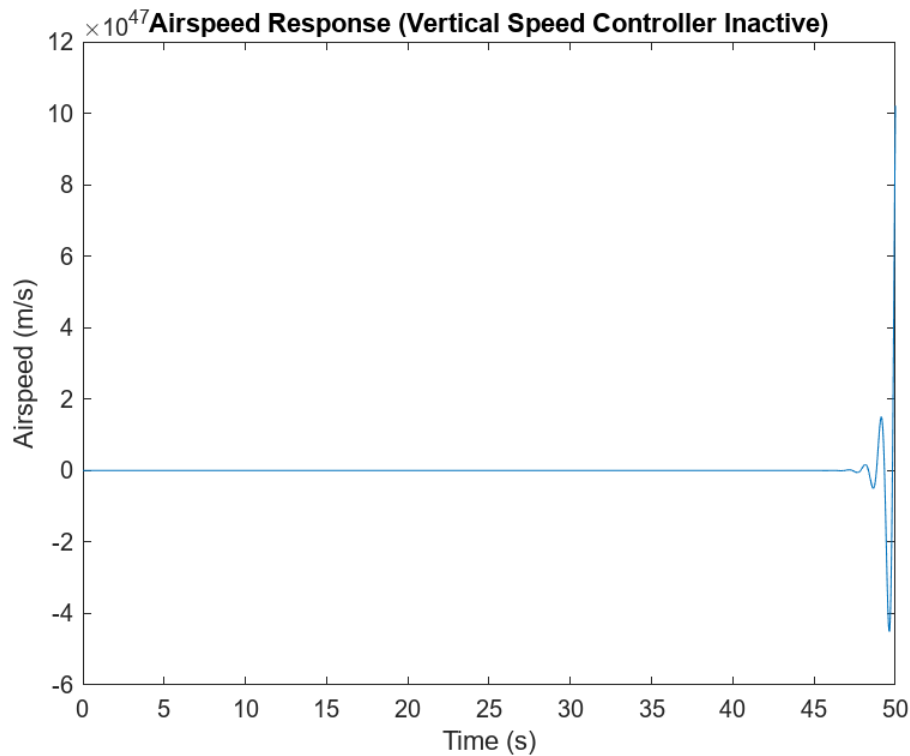


Figure 16: Throttle Control Activity (Auto Throttle System)

6. Final Design

6.1. Q5 – Implementation of the control and guidance laws

6.2. Q6 – Gust Effectiveness

6.3. Q7 – Commanded Large Vertical Speed

7. Conclusion

References

A. Example of a Table